

Structure and Function of the Autonomic Nervous System

Presented by Dr B.Benzouid

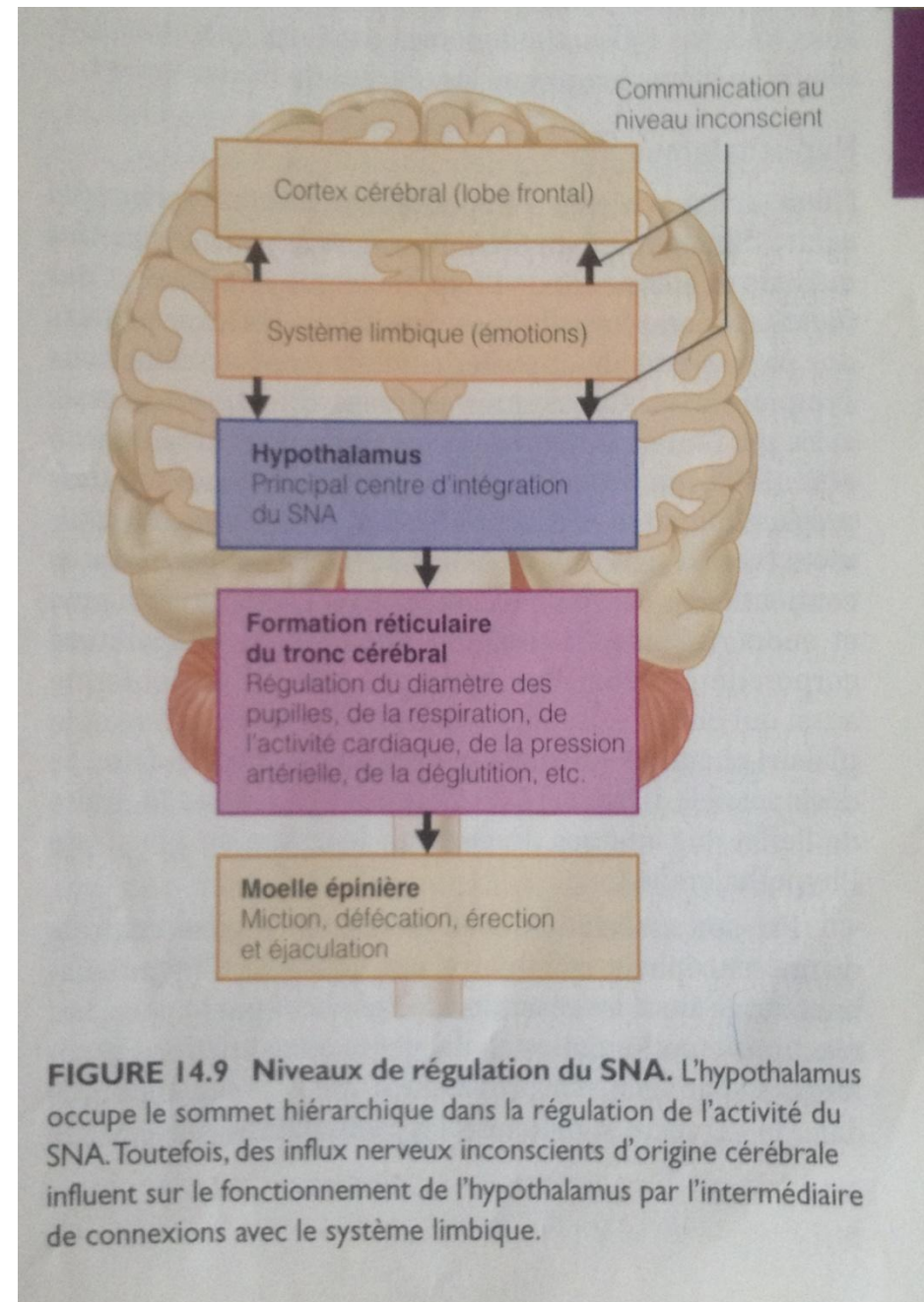
Faculty of Medicine, University of Constantine 3 Department of
Clinical Physiology and Functional Testing
Constantine University Hospital

Introduction

- ✓ The Autonomic Nervous System (ANS), also known as the Vegetative Nervous System (VNS), is part of the peripheral nervous system, which consists of the somatic nervous system and the VNS.
- ✓ Unlike the somatic nervous system, whose activities are under voluntary control and occur consciously, the VNS regulates internal organ functions, adapts them to current needs, and controls the body's so-called vegetative functions.
- ✓ Since these activities are beyond voluntary control, the VNS is also called the ANS. The ANS is an essential component in the processes of homeostasis regulation.

Introduction

- ✓ The ANS is involuntary, but its activity is regulated by the CNS through mechanisms located in the spinal cord, brainstem, hypothalamus, limbic system, and cerebral cortex.
- ✓ The hypothalamus serves as the central hub in the regulation of the ANS.



Introduction

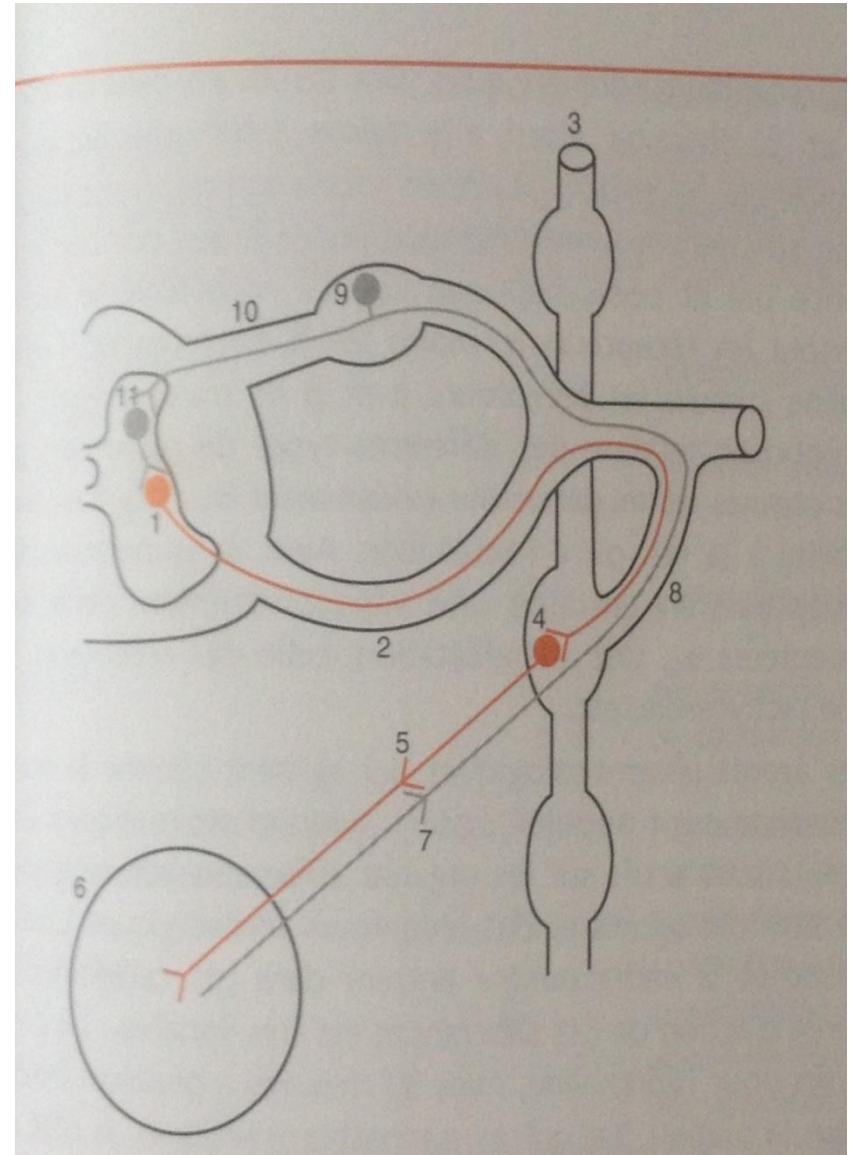
- ✓ The structure of the ANS reveals two poles that are both opposing and complementary, represented on the one hand by the sympathetic nervous system (SNS) and on the other by the parasympathetic nervous system (PNS).
- ✓ Each of these two systems has its own functional organization, regulatory centers, neurotransmitters, and receptors; However, despite these differences, there are constant complex interactions between these two systems, interactions that determine the sympathetic-vagal balance in each organ.
- ✓ The ANS, the primary reflex regulatory system, modulates most so-called autonomic functions

II General organisation of the ANS

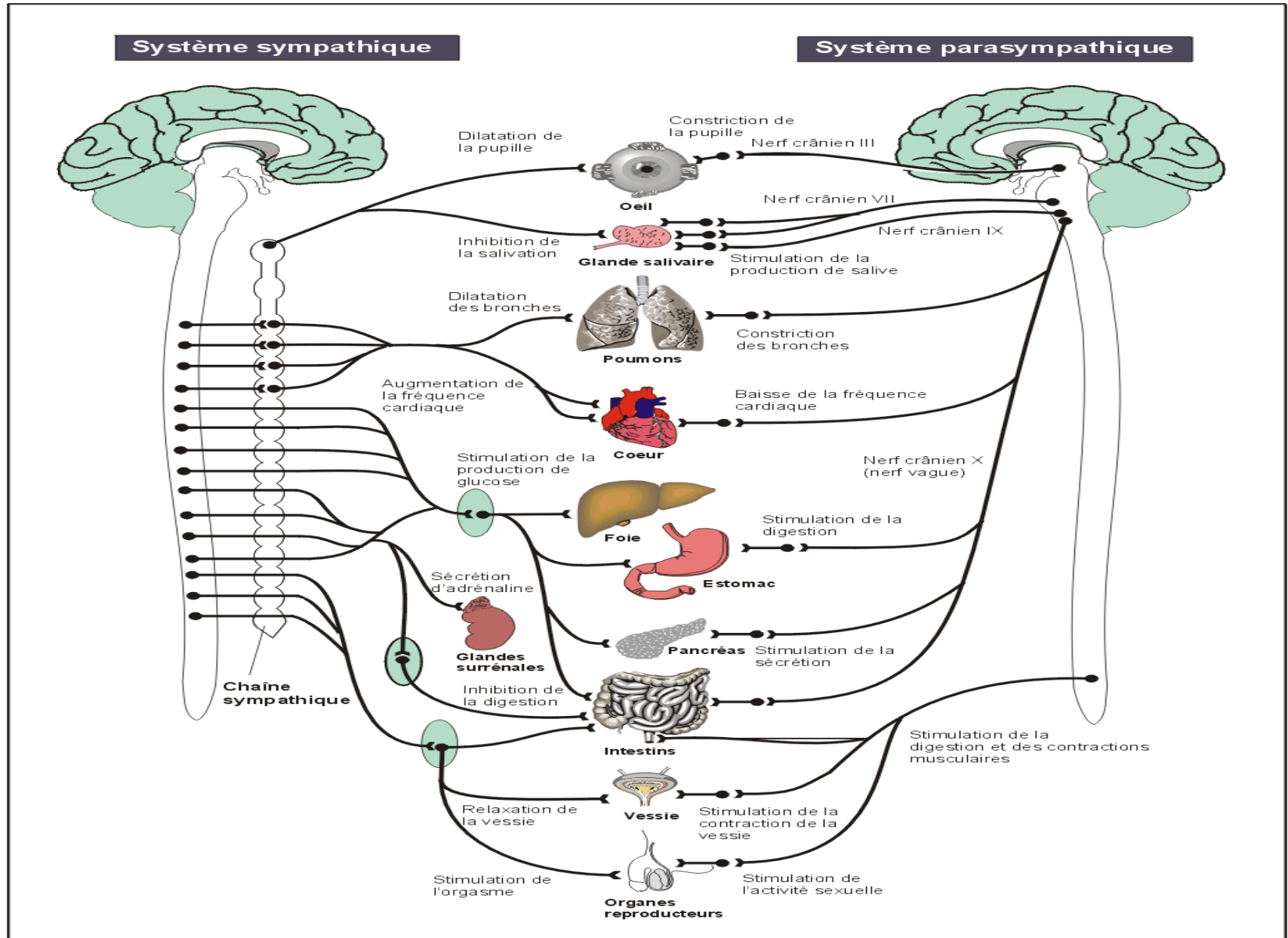
- ✓ The reflex arc, along with its afferent pathways, forms the functional basis of the autonomic nervous system (ANS).
- ✓ Afferent fibers carry information related to pain, as well as signals from mechanoreceptors and chemoreceptors in the vascular system, lungs, and digestive tract...
- ✓ Efferent fibers control the responses of the smooth muscles in various organs and the functioning of the heart and glands.
- ✓ The peripheral efferent pathways of the ANS consist of preganglionic fibers that form synapses with postganglionic fibers in the ganglia.

General organisation of the ANS

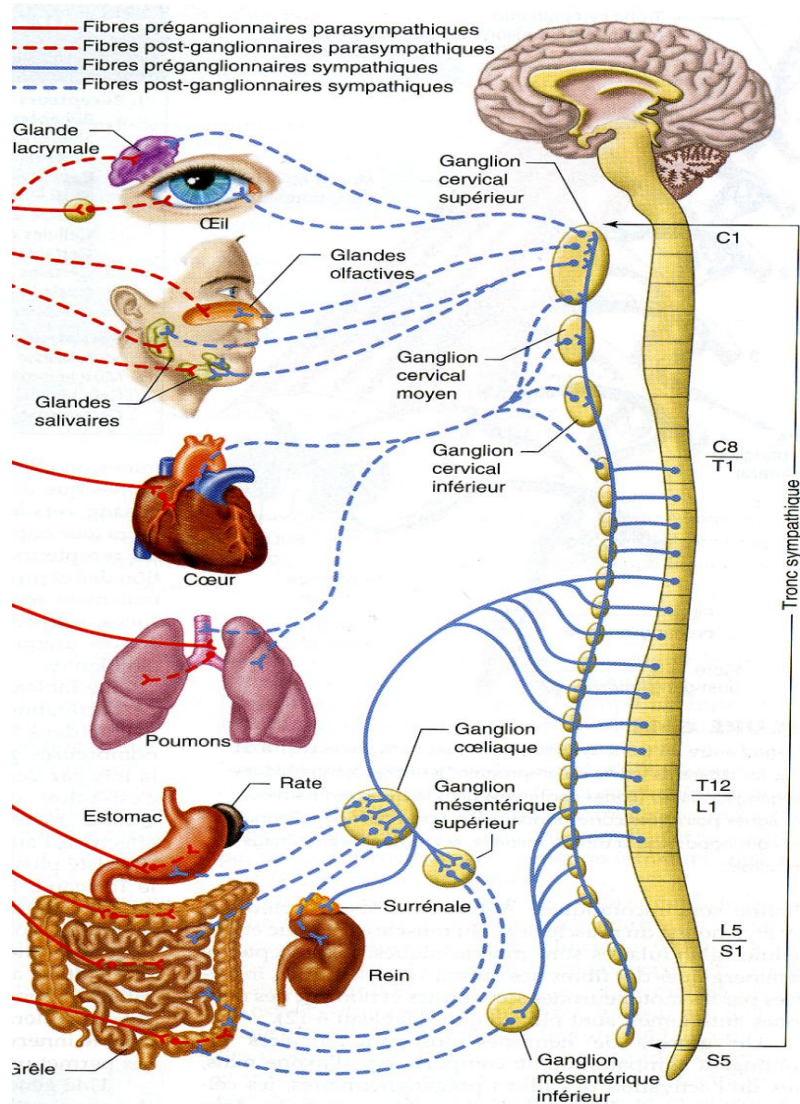
- ✓ Afferent information (7) travels through the white communicating branch (8) and the dorsal root (10) to terminate on a medullary interneuron.
- ✓ The effector neuron of the ANS is located in the intermediolateral column of the spinal cord (1). Its axon exits the spinal cord via a ventral root (2) and relays information in a ganglion of the paravertebral chain (3) to a second neuron (4), whose axon (5) innervates the target organ (6).
- ✓ The effector neuron receives information about the state of the target organ and can modulate its function via this reflex pathway



General organisation of the ANS

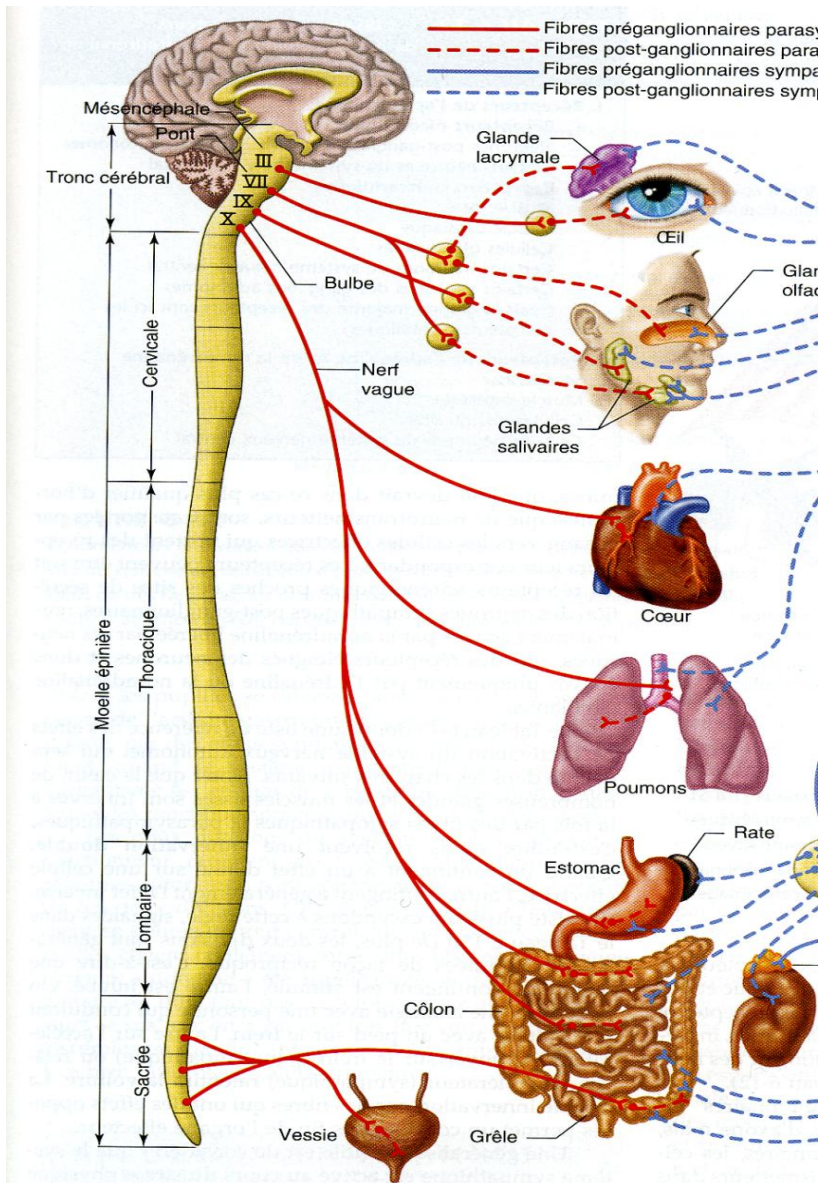


General organisation of the ANS

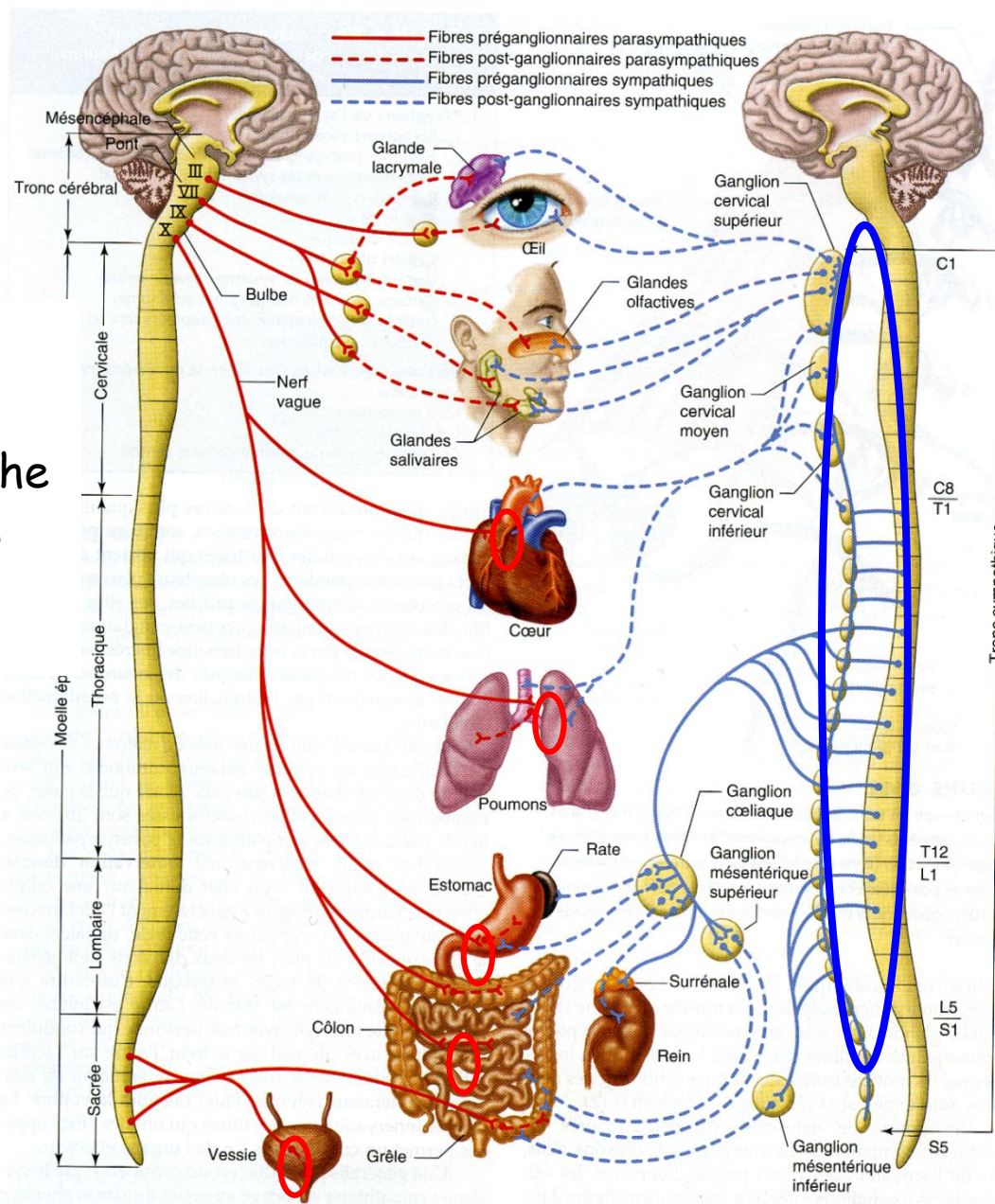


- SN Sympathetic: Nerves of the thoracic and lumbar spinal cord Active during emergencies (fight or flight)
- Neurotransmitter = **noradrenaline**

General organisation of the ANS



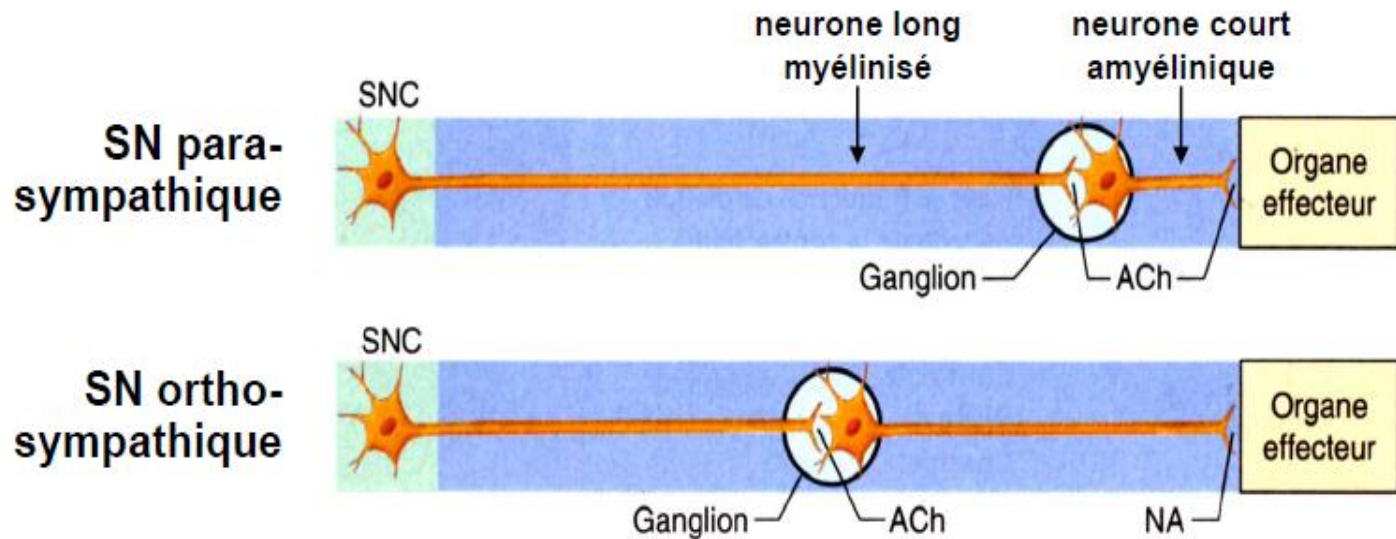
- SN Para Sympathetic:
Brainstem, cranial nerves (III, VII, IX, and X).
Nerves of the sacral region of the spinal cord.
- Active at rest
- Neurotransmitter = **Acetylcholine**



Parasympathetic
 ganglia
 are located **near** the
 organ or within
 the organ they
 innervate.

The
 sympathetic
 ganglia
 are arranged
 in a **chain**
 along the
 spinal cord

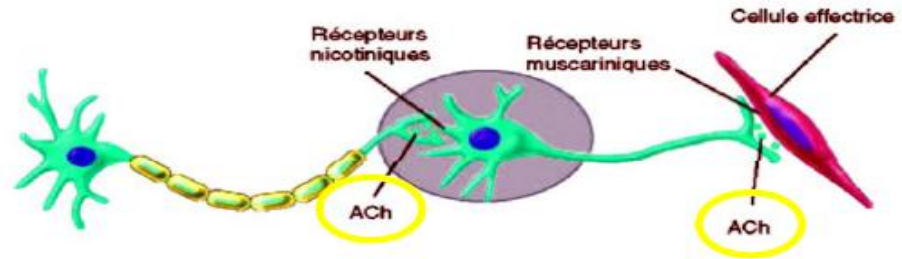
General organisation of the ANS



III- Synaptic Transmission of the ANS

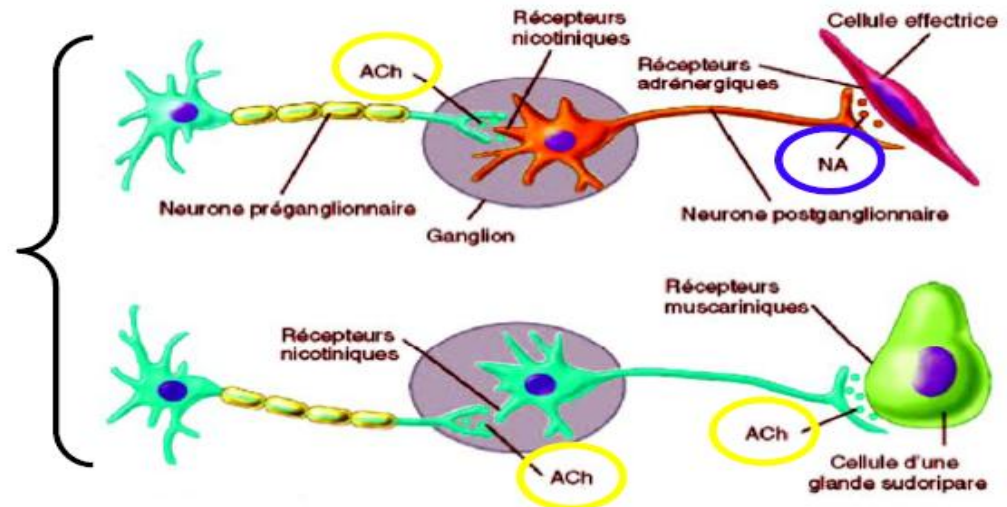
I

Système nerveux parasympathique



ACh = acétylcholine
NA = noradrénaline

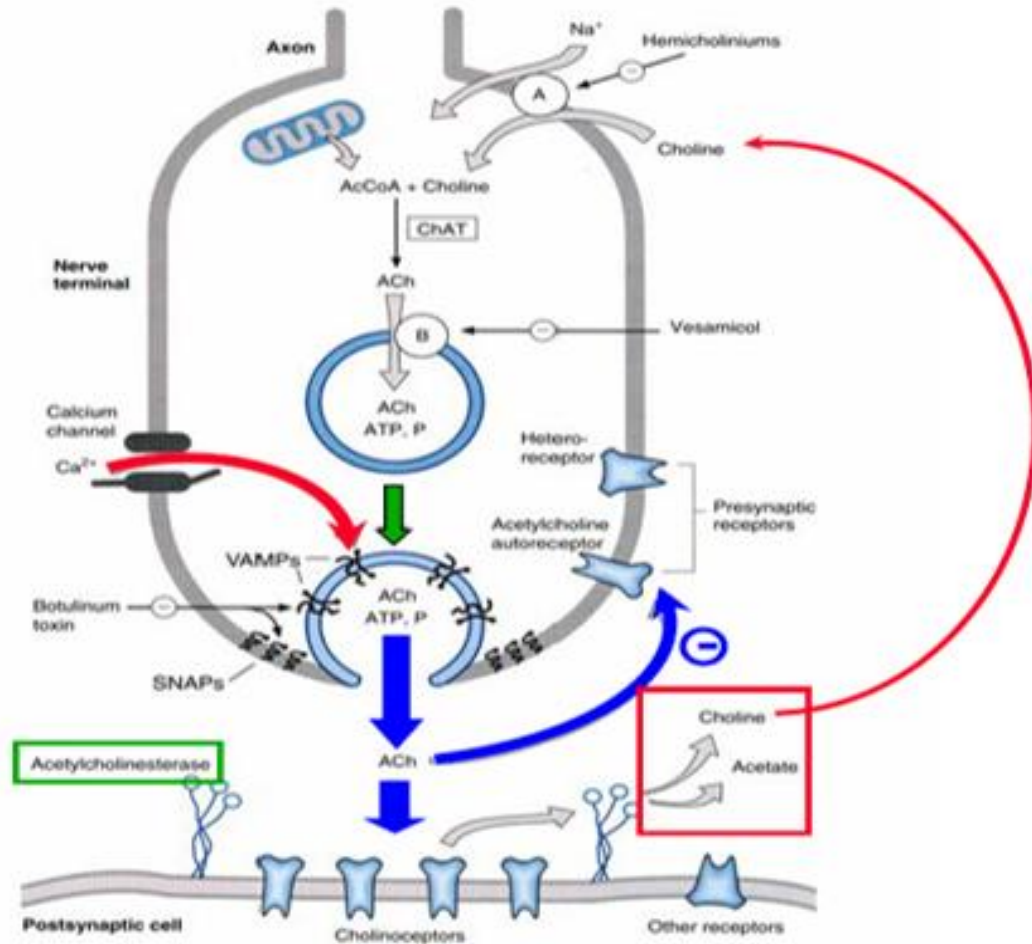
Système nerveux orthosympathique



Glande médullosurrénale



1-Cholinergic Transmission

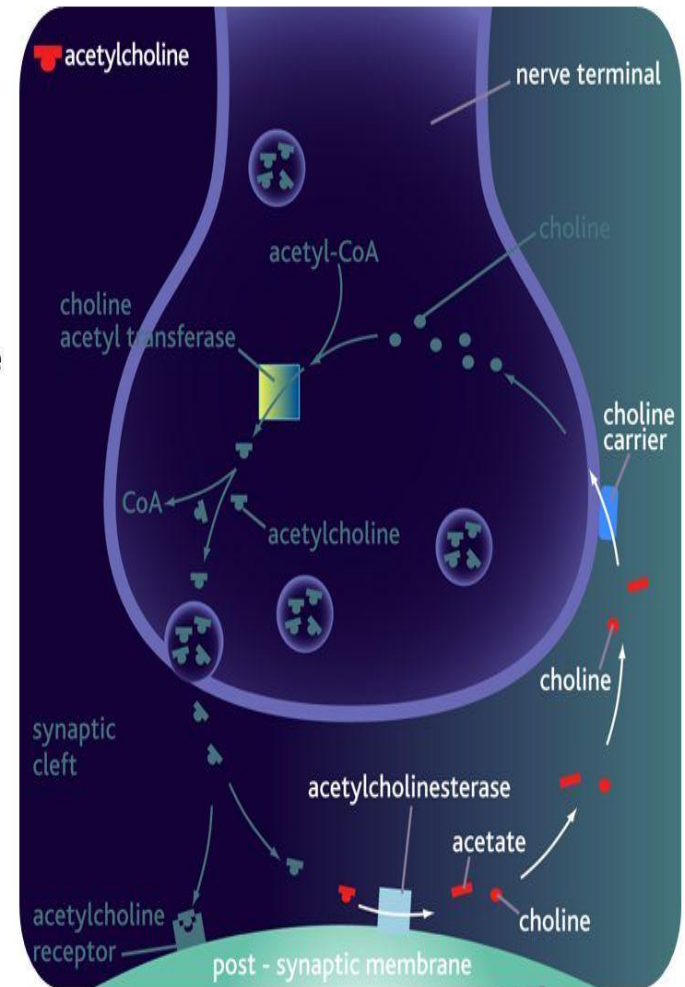


1-Cholinergic Transmission

- Neurotransmitter: Acetylcholine
Synthesis: Choline + Acetyl-CoA;
enzyme: choline acetyltransferase
- ✓ Acetylcholine is stored in vesicles.
- ✓ After release from the vesicles via exocytosis, its half-life is very short. It is broken down by the enzyme acetylcholinesterase.
- ✓ The choline resulting from this breakdown is reabsorbed (active choline uptake).
It is used for the synthesis of new acetylcholine.
- ✓ The action is very brief.

La Transmission Cholinergique

Figure n°2:
Synthèse et
métabolisme de
l'Ach



2 types of acetylcholine receptors

Spinal Cord

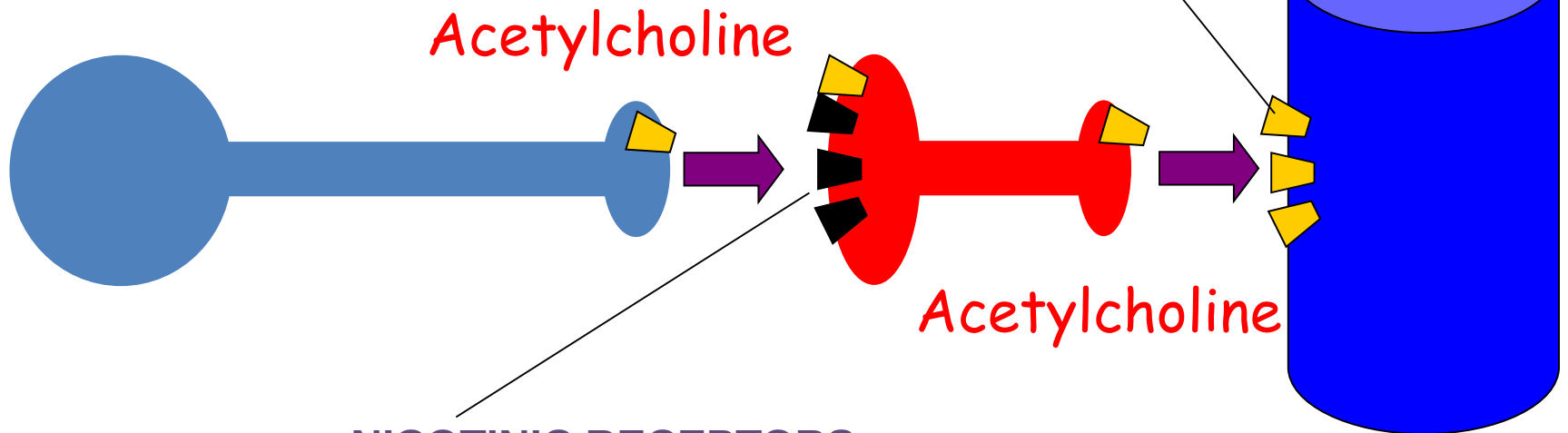
Ganglion

Target Tissue

MUSCARINIC receptors

Stimulated by muscarine

Pre- and post-synaptic localization on the target organ and ganglion relay



NICOTINIC RECEPTORS

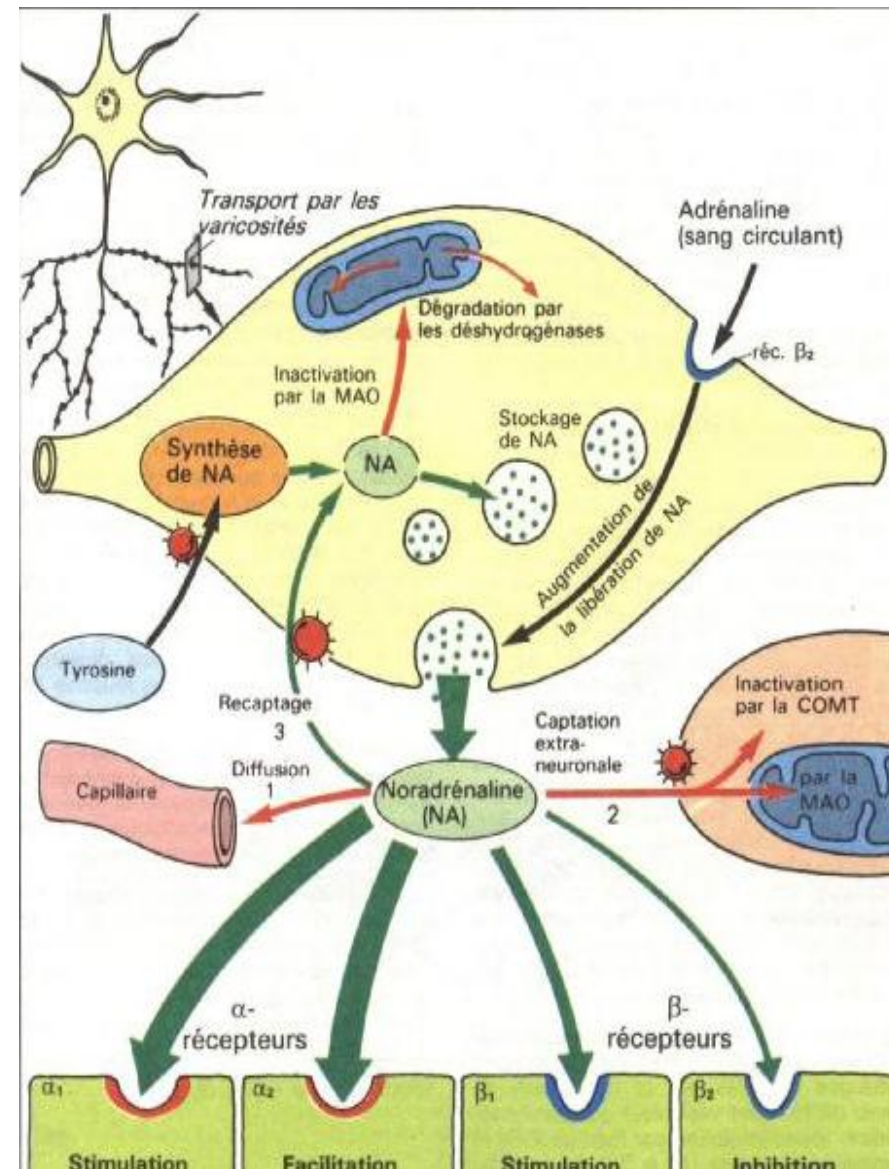
Stimulated by nicotine Located postsynaptically at the ganglionic relay

1-Cholinergic Transmission

- ✓ Several types of muscarinic receptors:
 - M1: Gastric acidity
 - M2 : Heart
 - M3: Lung, Eye, Bladder, Uterus.

2. Adrenergic Transmission

- ✓ The synthesis of **noradrenaline (NA)** is more complex.
- ✓ NA represents one of the final steps in the synthesis of **catecholamines**.
- ✓ The cell membrane of the **post-ganglionic sympathetic nerve fibers** bearing **synaptic vesicles** takes up, through a process of **active transport**, **L-Tyrosine**, the precursor of NA synthesis.



2. Adrenergic Transmission

- Tyrosine is converted into L-DOPA by tyrosine hydroxylase.
- This enzymatic reaction is accelerated by Na^+ and Ca^{++} and inhibited by L-DOPA (negative feedback).
- L-DOPA is converted into dopamine by DOPA decarboxylase and then into Noradrenaline through hydroxylation.

METABOLISME DE LA (NOR)ADRENALINE

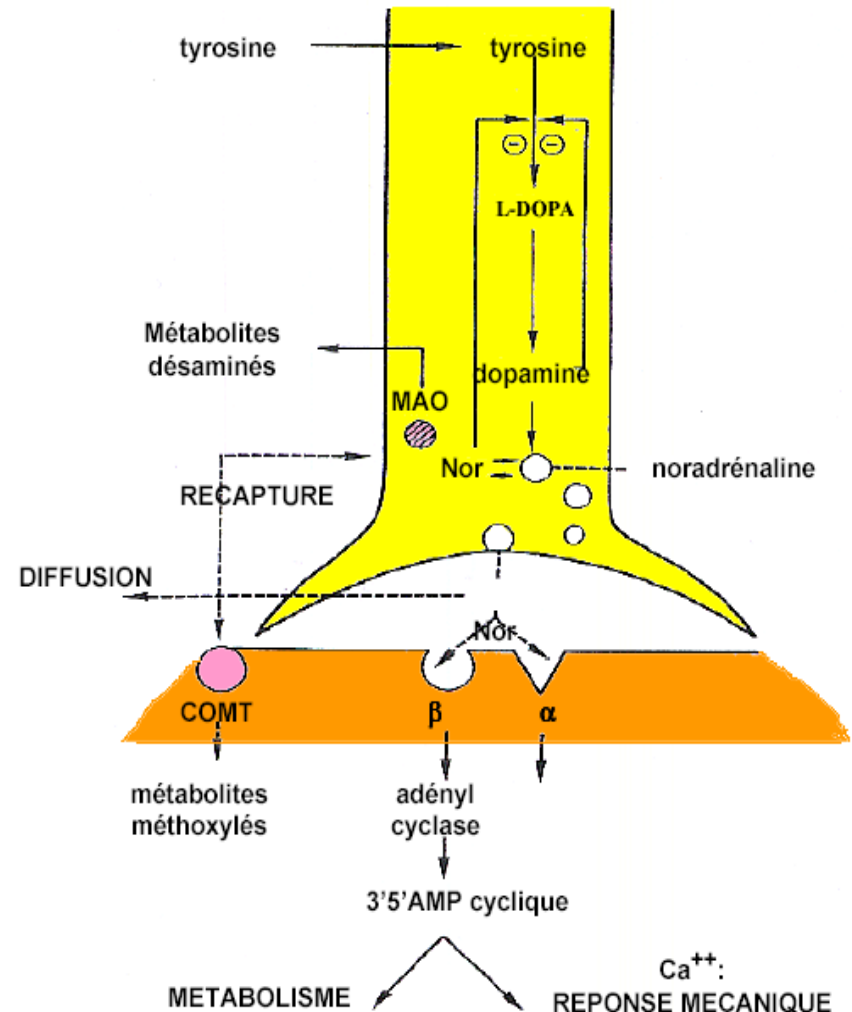
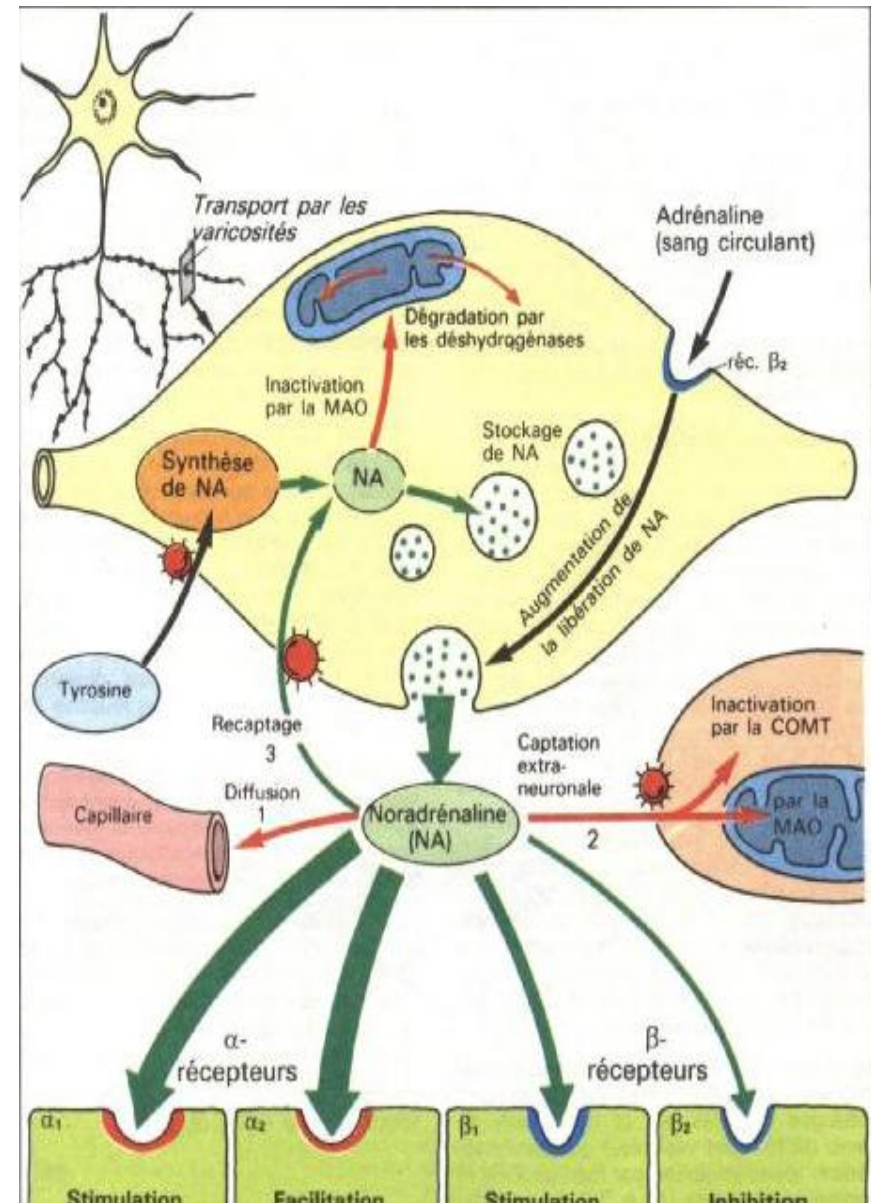


Fig. 5 : Schéma d'une jonction noradrénergique.

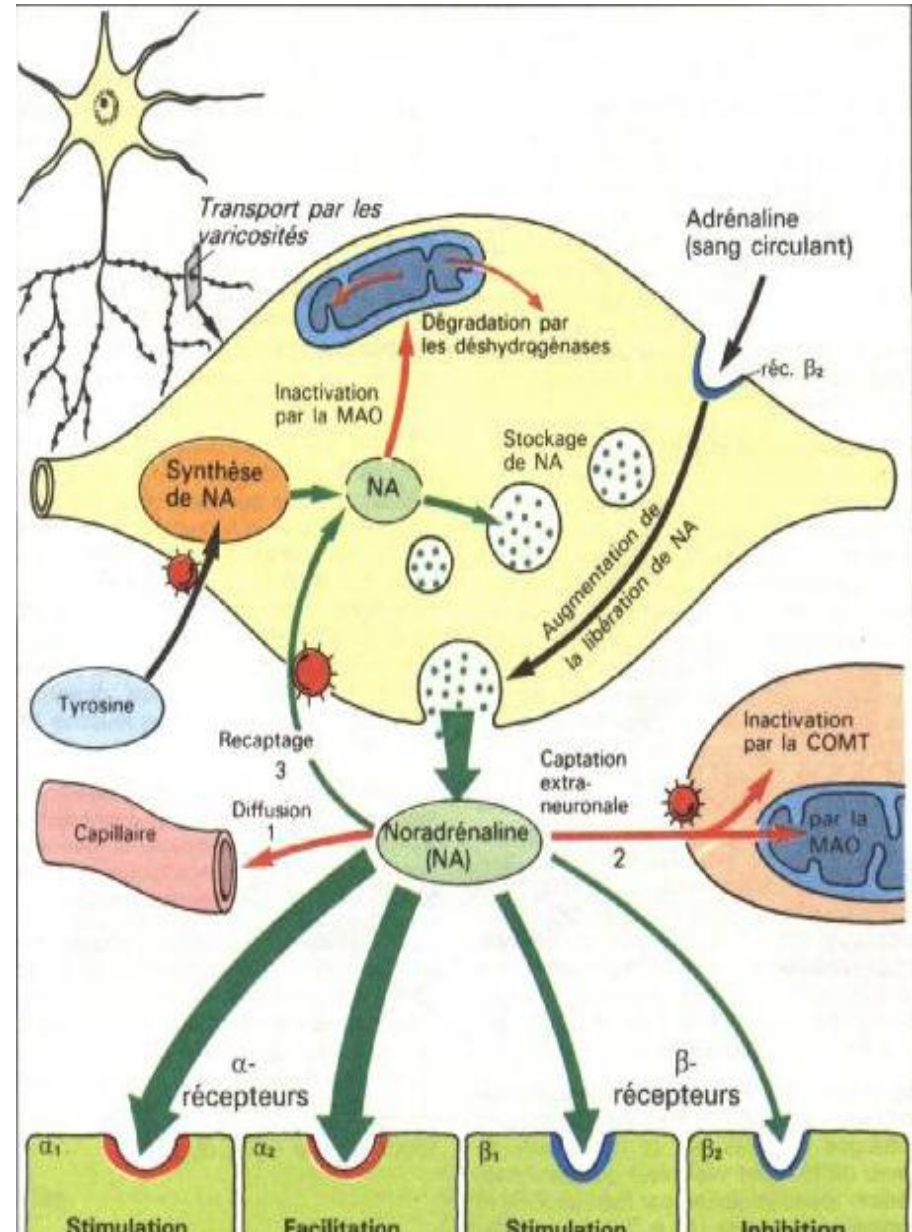
Adrenergic Transmission

- NA is stored in small granular vesicles in the form of a micellar complex.
- NA is released when an action potential reaches the synapse.
- ✓ The arrival of this AP triggers a calcium current: an important role. However, the mechanism of NA exocytosis remains unknown.
- ✓ The methylation of NA results in adrenaline.



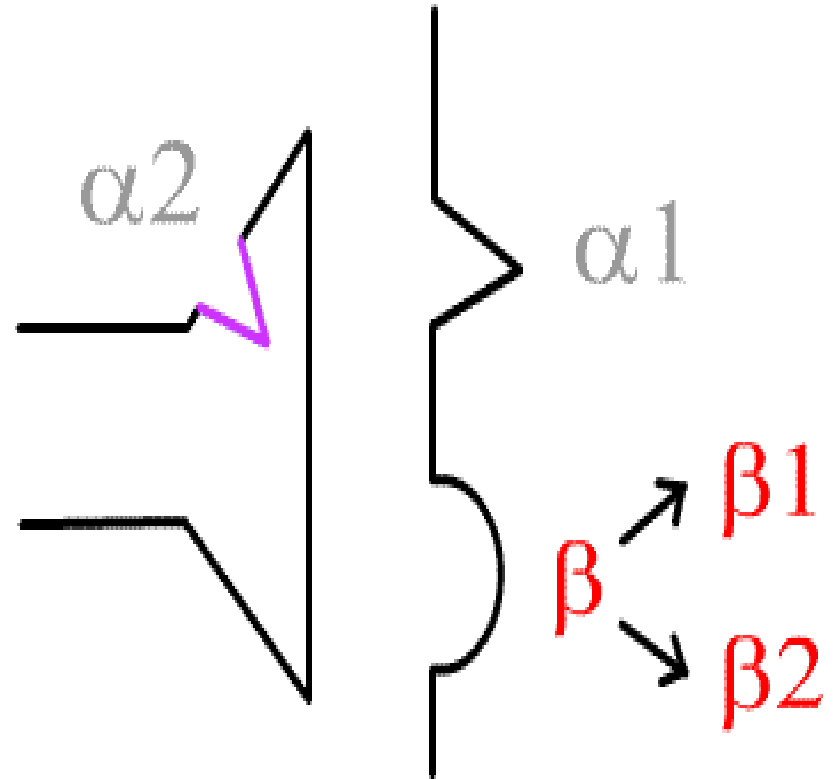
Adrenergic Transmission

- ✓ Once released, NA is eliminated through three mechanisms, listed in descending order of importance
 - Presynaptic reuptake (70%): free NA is inactivated by mitochondrial MAO
 - Diffusion: synaptic cleft → blood
 - Enzymatic degradation by MAO and COMT.
- ✓ NA secreted into target organs, like acetylcholine, has a short half-life.
- ✓ In contrast, NA and adrenaline released from the adrenal medulla enter the circulation and exert their effects for a longer duration (10–30 s).



Adrenergic Transmission

- 2 types of α receptors: α_1 , α_2
- ✓ α_1 receptor: Postsynaptic only
Smooth muscle: Arterioles, Uterus, Bronchioles, Salivary glands.
- ✓ α_2 receptor : Presynaptic and postsynaptic
- ✓ CNS, Kidneys, Uterus,

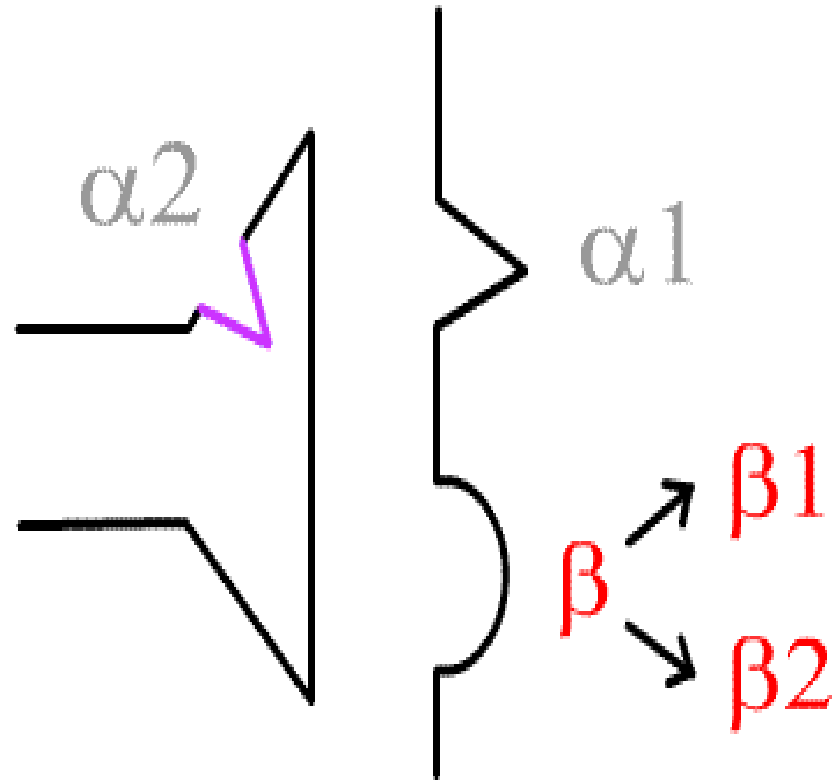


Adrenergic Transmission

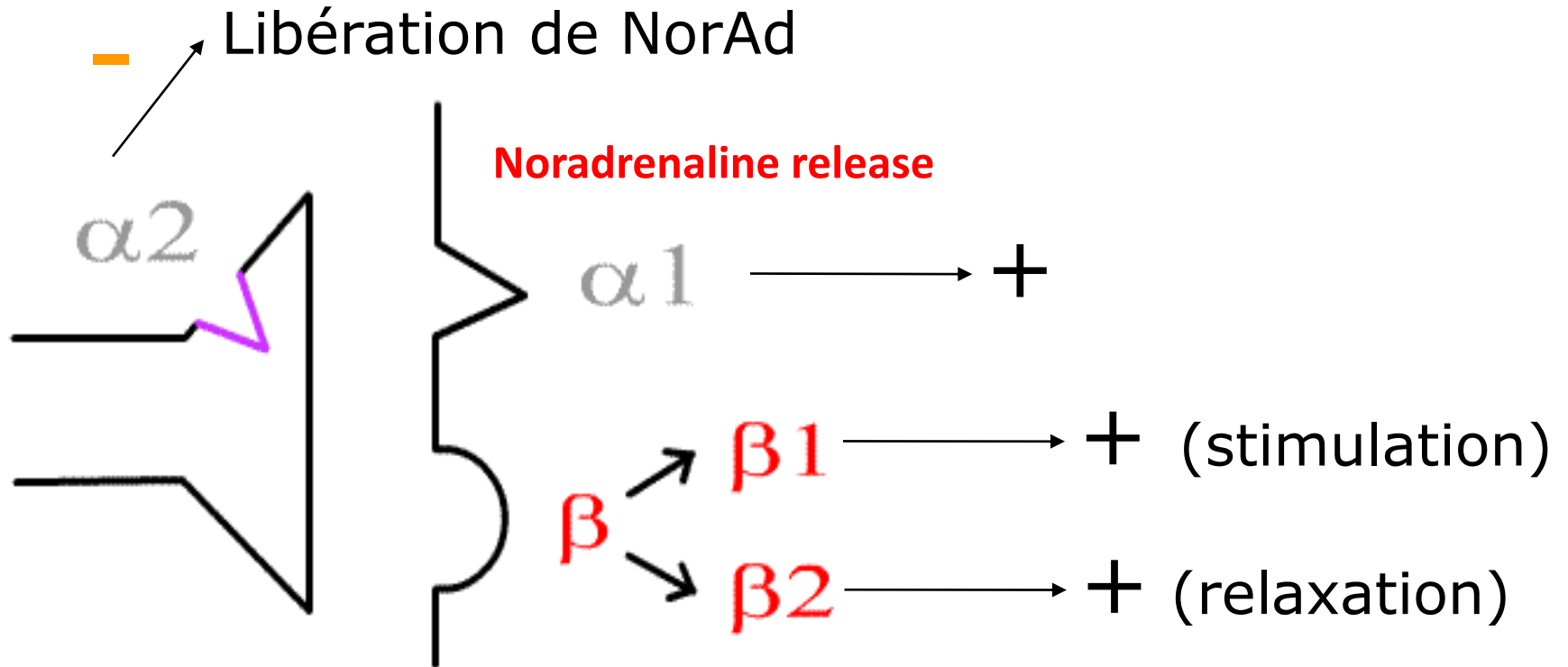
- 2 types of β receptors: β_1 and β_2

β_1 : Heart
(dromotropic, chronotropic, inotropic +)
Kidney
(increased renin secretion)

✓ β_2 : Lungs, Coronary arteries



Adrenergic Transmission: Noradrenaline Receptors



Adrenergic Transmission

- ✓ All these receptors are activated through a **G protein**.
- ✓ They use as a second messenger:
 - **$\alpha 1$** : the **IP3 / DAG** pathway
 - **$\alpha 2$** : the **cAMP** pathway
 - **β** : the **cAMP** pathway

Adrenal medulla :

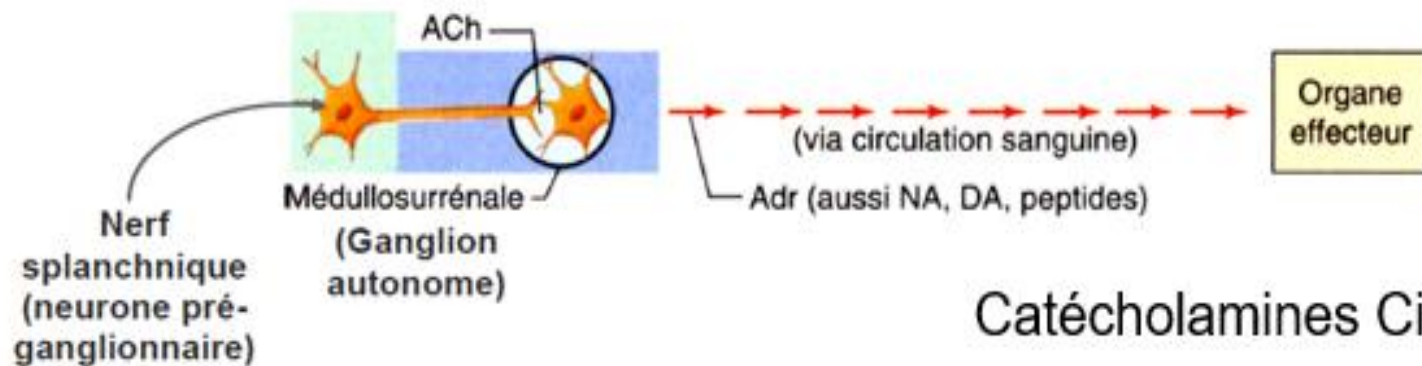
Cellules chromaffines

- Adrénaline (80%) > Noradrénaline (20%)
- formation séparée / innervation distincte / stimulus préférentiel

Hypoglycémie → Adrénaline

Asphyxie → Noradrénaline

- catécholamines = situations d'urgence, "flight or fight" (fuite ou combat)



Catécholamines Circulantes

- médullo-SRN et SN sympathique = système sympatho-adrénergique
- indépendance / mise en jeu préférentielle

SNS = cardiovasculaire
SRN = métabolisme

IV - Non-adrenergic, non-cholinergic system

- ✓ A third system has been identified: **NANC**
 - **NANCe** (bronchoconstrictive neuropeptides: **Tachykinin, Neurokinins A and B, Substance P**)
 - **NANCI** (VIP...)

V - Effects of the Parasympathetic ANS on some organs

œil sphincter irien muscle ciliaire	mydriase = myosis contraction = accomodation	M3
tube digestif muscle lisse sphincters	contraction relâchement	M3
glandes exocrines	sécrétion	M3
Pression artérielle	baisse	
cœur	bradycardie sinusale	M2
endothélium vasculaire	libération de NO	M3
bronches muscle lisse	contraction	M3
utérus	contraction relaxation (utérus gravide)	M3
vessie détrusor sphincter	contraction relaxation	M3
org. génitaux mâles	érection	
peau	sudation	

VI - Effects of the sympathetic ANS on some organs

œil muscle radié iris muscle ciliaire	$\alpha 1$	$\beta 2$	mydriase relaxation
tube digestif muscle lisse sphincters	$\alpha 1$ $\alpha 1$	$\alpha 2$ $\beta 1$ $\beta 2$	relaxation contraction
pancréas endocrine	$\alpha 2$	$\beta 2$	baisse sécrétion augmentation sécrétion +++
foie	$\alpha 1$	$\beta 2$	glycogénolyse gluconéogénèse
cœur		$\beta 1$ $\beta 2$	effet inotrope effet chronotrope, dromotrope
artères coronaires	$\alpha 1$	$\beta 2$	constriction dilatation +++
cutanées	$\alpha 1$		constriction+++
cérébrales	$\alpha 1$	$\beta 2$	constriction dilatation
pulmonaires	$\alpha 1$	$\beta 2$	constriction dilatation
rénales	$\alpha 1$	$\beta 2$	constriction dilatation
veines	$\alpha 1$	$\beta 2$	constriction relaxation
bronches muscle lisse		$\beta 2$	relaxation
reins		$\beta 1$	sécrétion de rénine
vessie détrusor sphincter	$\alpha 1$	$\beta 2$	relaxation contraction
utérus	$\alpha 1$	$\beta 2$	contraction relaxation (utérus gravide)
org. génitaux mâles	$\alpha 1$		éjaculation
peau	$\alpha 1$		piloérection, sudation
adipocytes		$\beta 3$	lipolyse, thermogénèse

VII. Pharmacology of the Autonomic Nervous System

Sympathomimetics

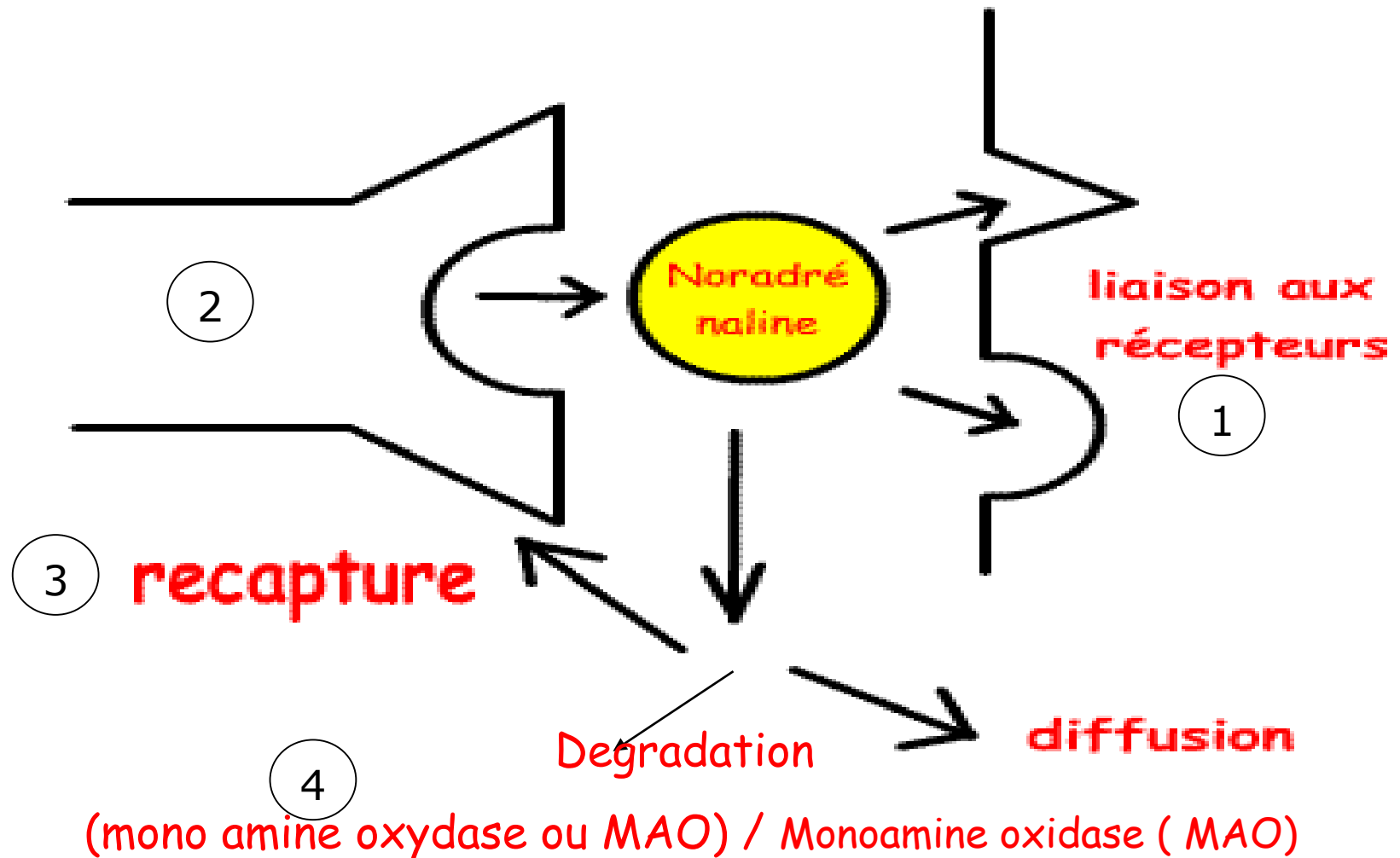
Sympatholytics

Parasympathomimetics

Parasympatholytics

Ganglion blockers

Pharmacological agent of the Sympathetic system

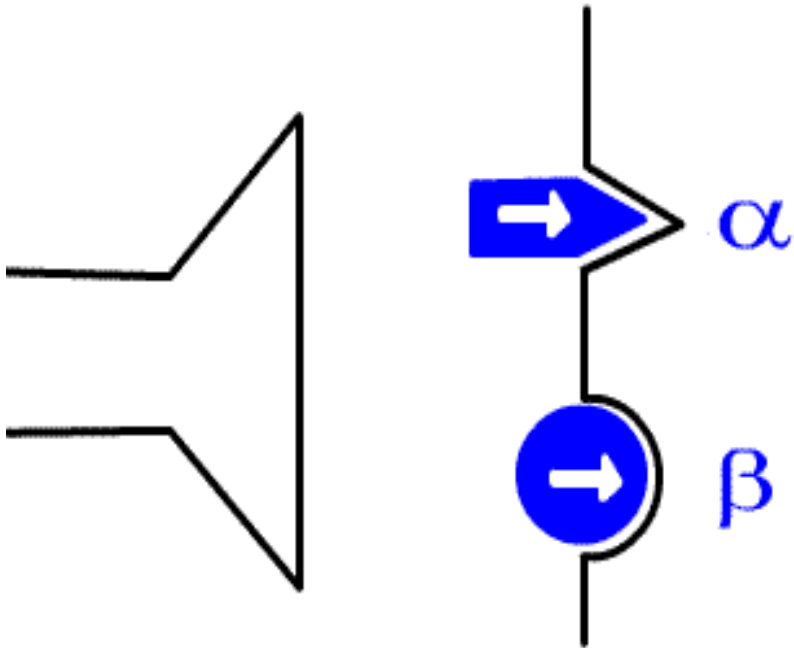


1 . Acting on receptors (agonists, antagonists)

2. Acting on the synthesis, storage, and release of Noradrenaline

3 and 4. Preventing the reuptake and degradation of Noradrenaline

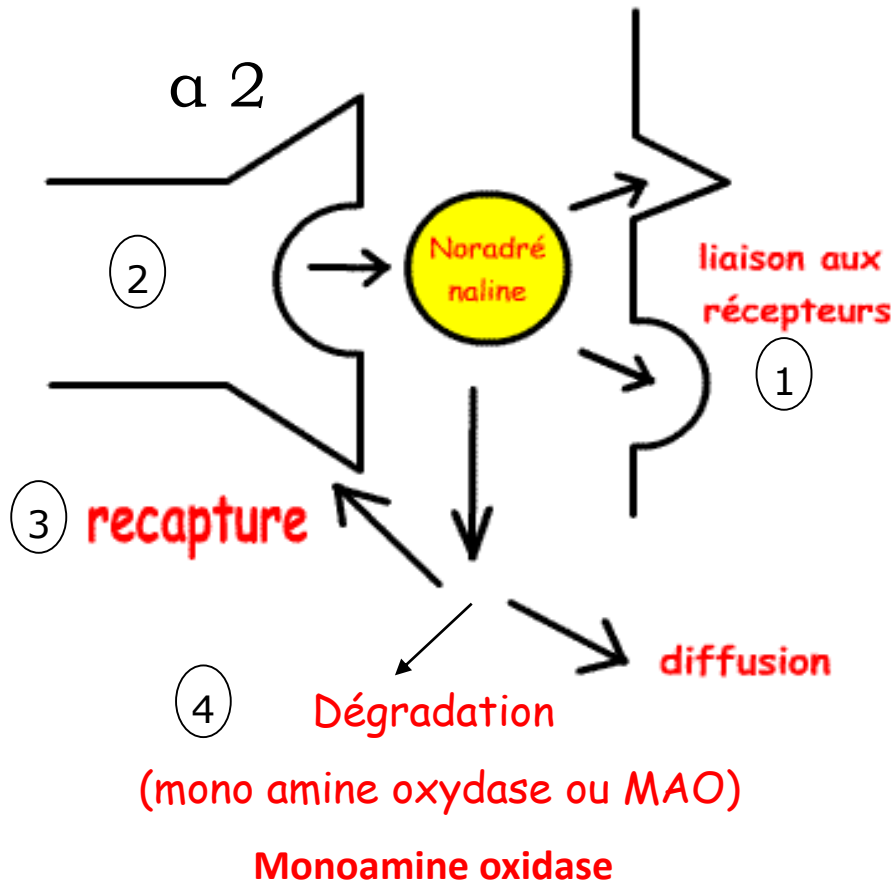
Direct Sympathomimetics



Alpha-1 agonists: adrenaline, noradrenaline, dopamine, phenylephrine

Beta-1 agonists: isoprenaline, dobutamine, dopamine, salbutamol, salmeterol, formoterol, fenoterol, clenbuterol, orciprenaline, terbutaline

Indirect Sympathomimetics : 2, 3, 4 — Increase of the neuromediator : sympathoplegia, antiadrenergic.



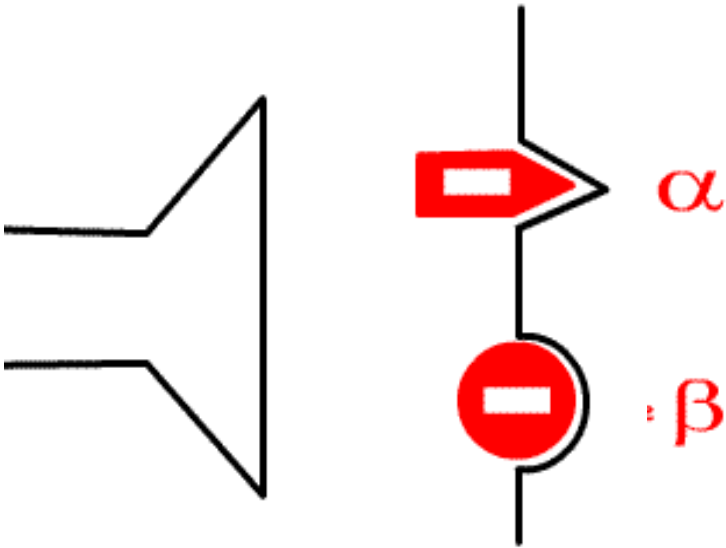
Alpha-2 antagonists
(moderate the inhibitory effect of alpha-2 on noradrenergic synthesis):
yohimbine, idazoxane

Reuptake inhibitors: imipramine, cocaine « 3 »

MAOIs: phenelzine, pargyline, selegiline, toloxatone « 4 »

Noradrenaline release stimulants « 2 »: amphetamine, ephedrine, tyramine

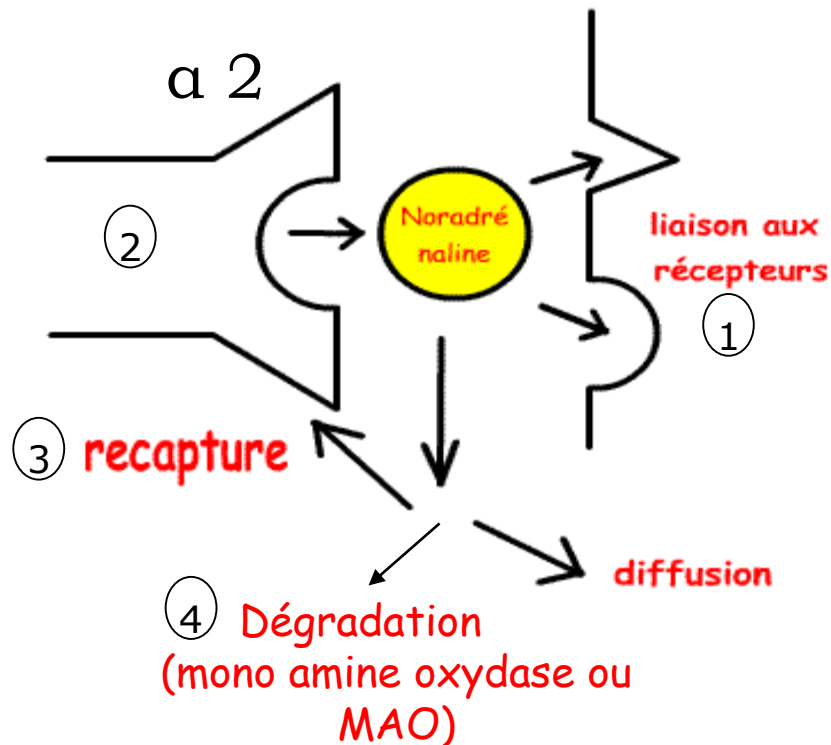
Direct Sympatholytics : « 1 » Antagonists, Adrenolytics



Alpha-1 antagonists : carvedilol, prazosin, doxazosin, terazosin, phenoxybenzamine, phentolamine, labetalol

Beta-1 antagonists : propranolol, atenolol, metoprolol, alprenolol, labetalol

Indirect Sympatholytics : 2, 3, 4 — Decrease of the neuromediator :
sympathoplegia, antiadrenergic

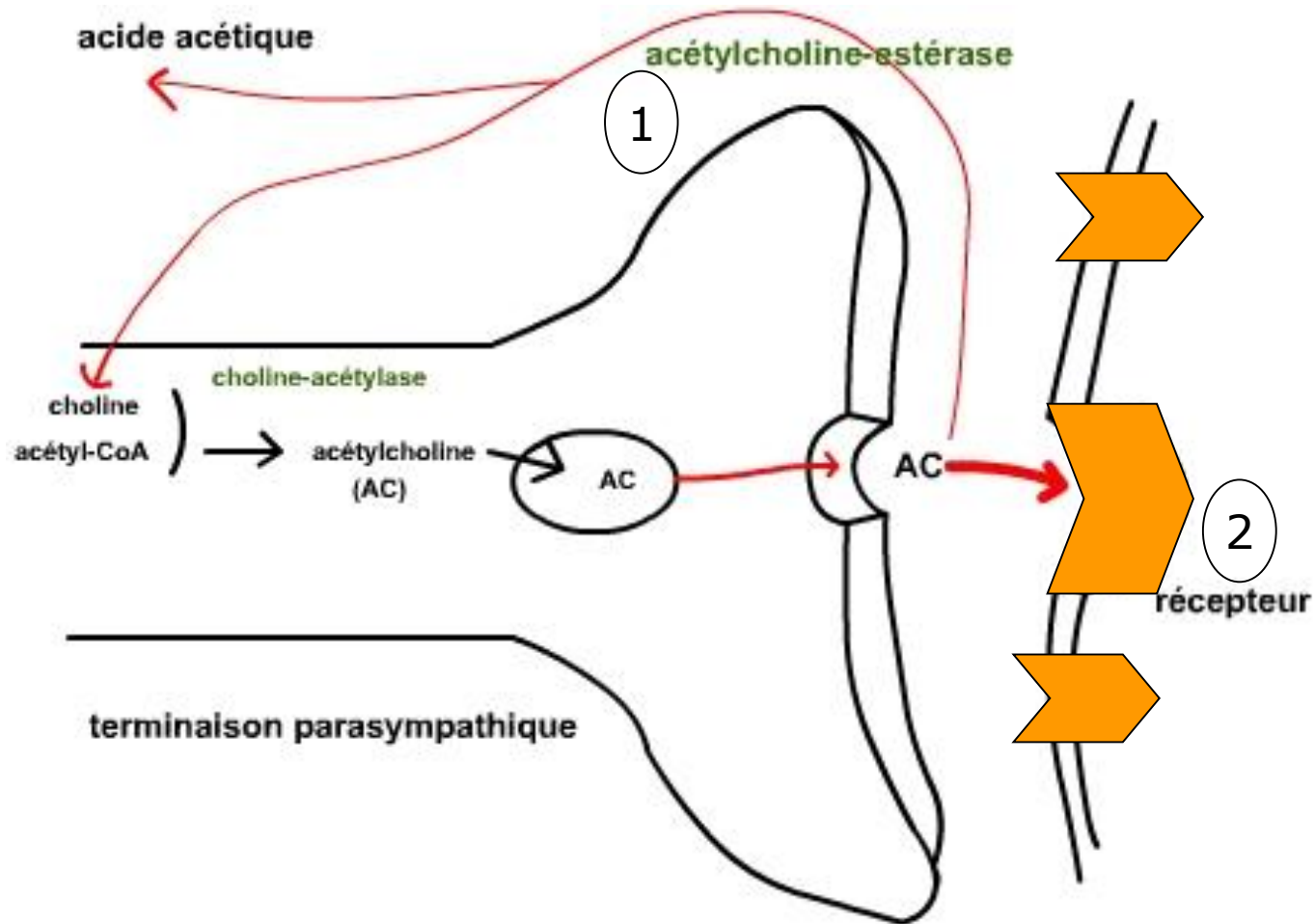


Alpha-2 agonists (*enhance the inhibitory effect on mediator release*) : clonidine, phenoxybenzamine, phentolamine, methoxamine

Noradrenaline release inhibitors : guanethidine

Granular storage inhibitors : reserpine

Pharmacological agent of the Parasympathetic system



- Preventing the degradation of acetylcholine
- Acting on acetylcholine receptors (agonists, antagonists)

Parasympathomimetics (+):

- **Direct** (cholinergic) “2” Muscarinic:
pilocarpine, carbachol, methacholine,
bethanechol, oxotremorine Nicotinic:
nicotine, lobeline

- **Indirect-acting agents “1”**

Anticholinesterases : prostigmine,
pyridostigmine, physostigmine,
edrophonium

Parasympatholytics (-)

Indirect: Nicotinic antagonists

Direct: Muscarinic antagonists
“atropine-like”

Atropine, ipratropium, tropicamide,
clopentolate, scopolamine,

Ganglioplegics

- ✓ Ganglionic transmission can be blocked by agents called ganglioplegics, such as:
 - **Trimetaphan camsylate** : Anticholinergic inhibition of both sympathetic and parasympathetic effects
 - **Hexamethonium** : Nicotinic anticholinergic

VIII-Conclusion

- ✓ Distributed throughout most of the body's organs and fostering coordination among them, the ANS plays a role in modulating and regulating unconscious vegetative functions
- ✓ While the organs and particularly the viscera each have their own autonomy, the ANS ensures their harmonious functioning while respecting their independence.
- ✓ If its action is interrupted, the organs survive and continue to function, but their activity is no longer organized in terms of homeostasis and response to stressors.
- ✓ The activities of the ANS extend to the entire organism and find practical applications, particularly in pathology and therapeutics.

Bibliographic Ref

- H.Guénard Physiologie humaine
- Guyton Précis de physiologie Humaine
- Elaine N. MARIEB Anatomie et physiologie Humaines